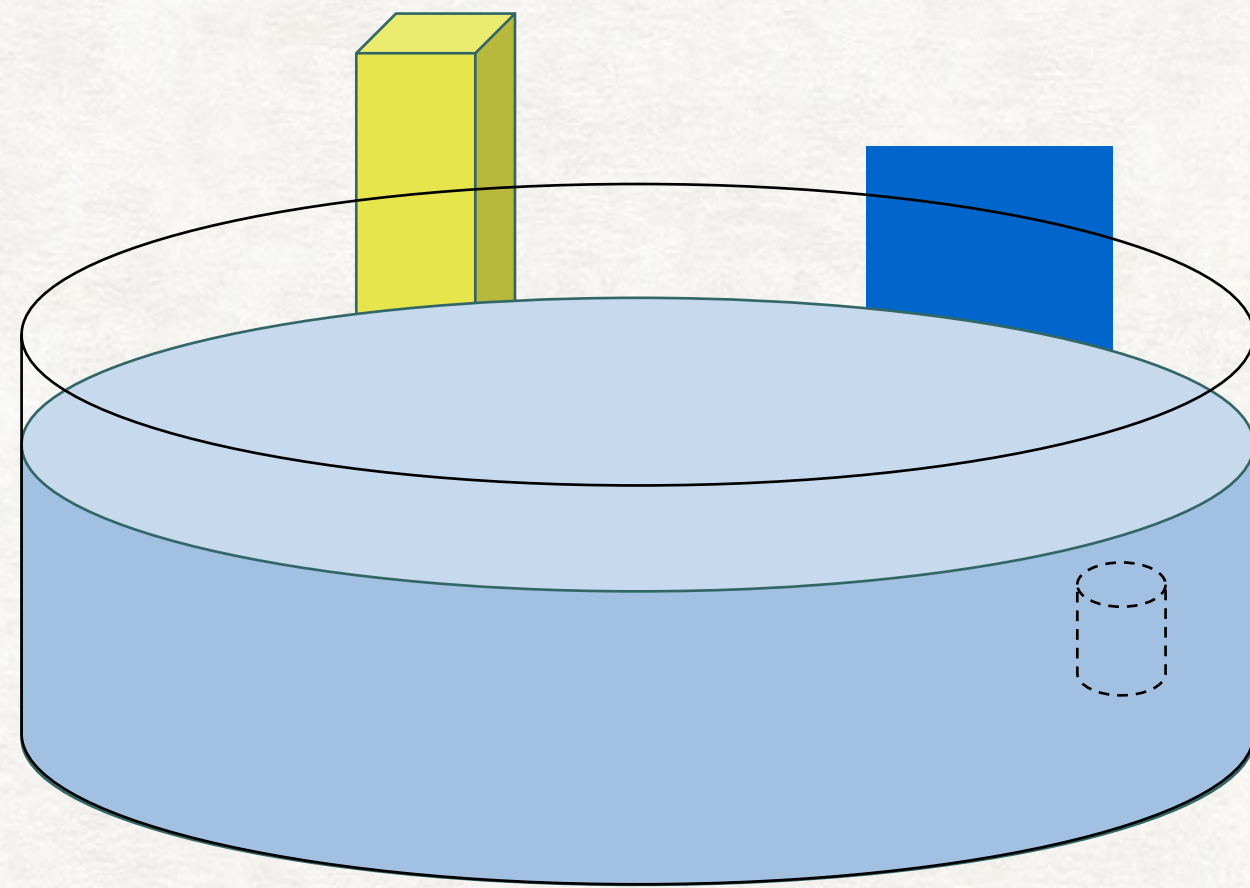
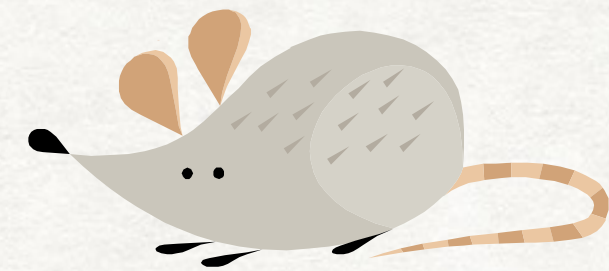


COM 3240

REINFORCEMENT LEARNING

MODELLING EXPERIMENTS WITH REINFORCEMENT LEARNING

THE MORRIS WATERMAZE



- Reinforcement Learning
 - Reward/Punishment
 - Exploration/Exploitation

Foster et al, 2000
Arleo et al, 2001
Stroesslin et al, 2005

MODELLING EXPERIMENTS WITH REINFORCEMENT LEARNING

TALK OUTLINE

- “Vanilla” Temporal Difference Learning
- Learning from the perspective of neuroscience
- Spike-based Reinforcement Learning of rodent navigation
- Tutorial I : IF model with escape noise
- Tutorial II : Reward Maximization rule
- Results
- Take-home message

MODELLING EXPERIMENTS WITH REINFORCEMENT LEARNING

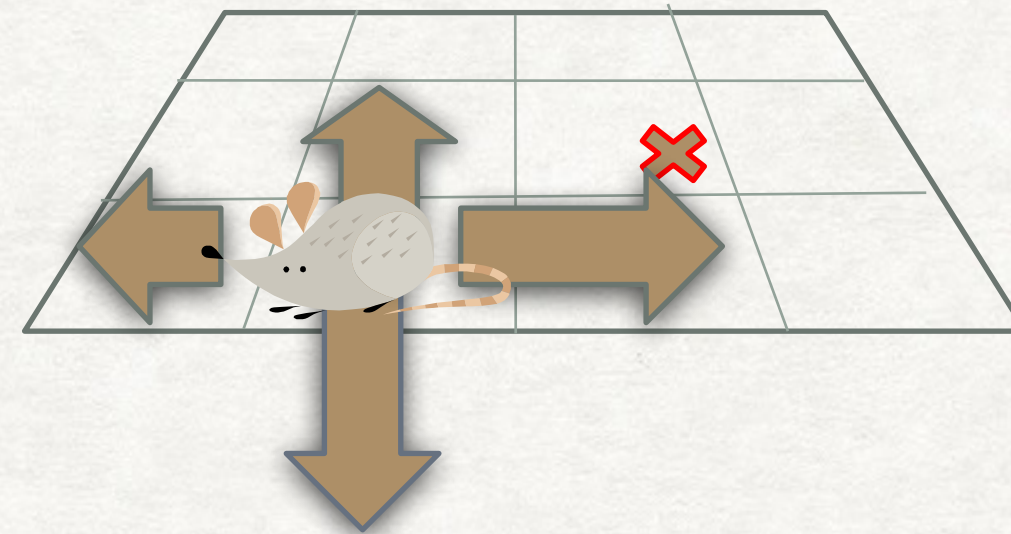
TALK OUTLINE

- **“Vanilla” Temporal Difference Learning**
- Learning from the perspective of neuroscience
- Spike-based Reinforcement Learning of rodent navigation
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MODELLING EXPERIMENTS

"VANILLA" TEMPORAL DIFFERENCE LEARNING

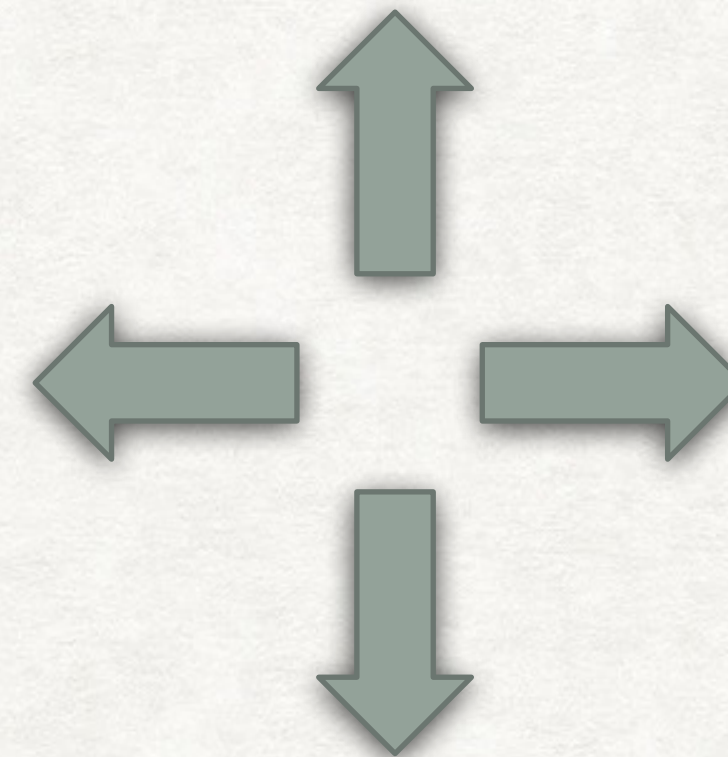
States x Actions: $3 \times 4 \times 4 = 48$



Maximize expected return

$Q(\text{state}, \text{action})$

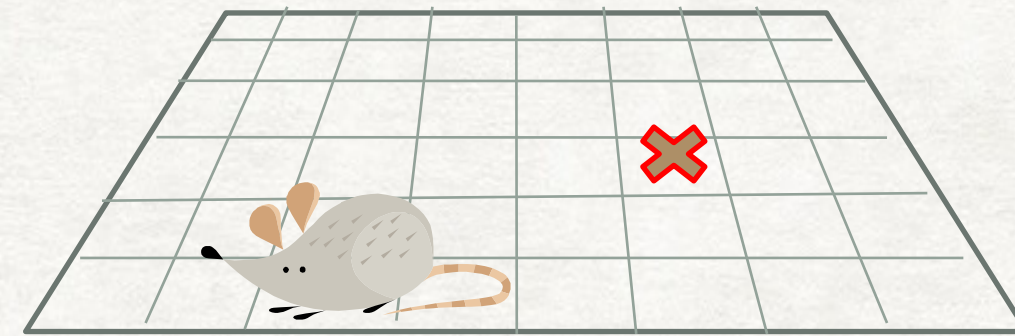
We do not know the Q values



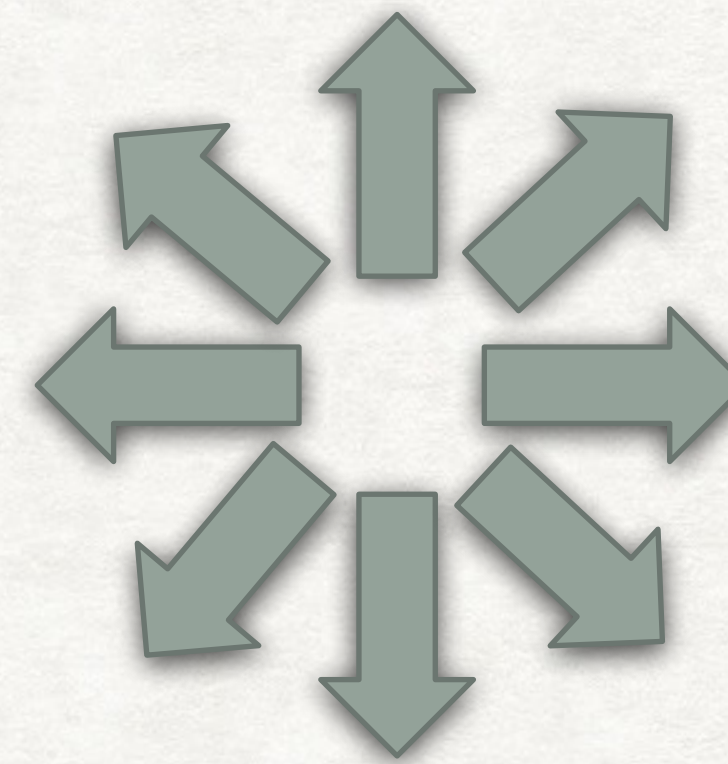
MODELLING EXPERIMENTS

"VANILLA" TEMPORAL DIFFERENCE LEARNING

States x Actions: $6 \times 8 \times 8 = 384$



Q (state, action)



Learning process slows down

MODELLING EXPERIMENTS WITH REINFORCEMENT LEARNING

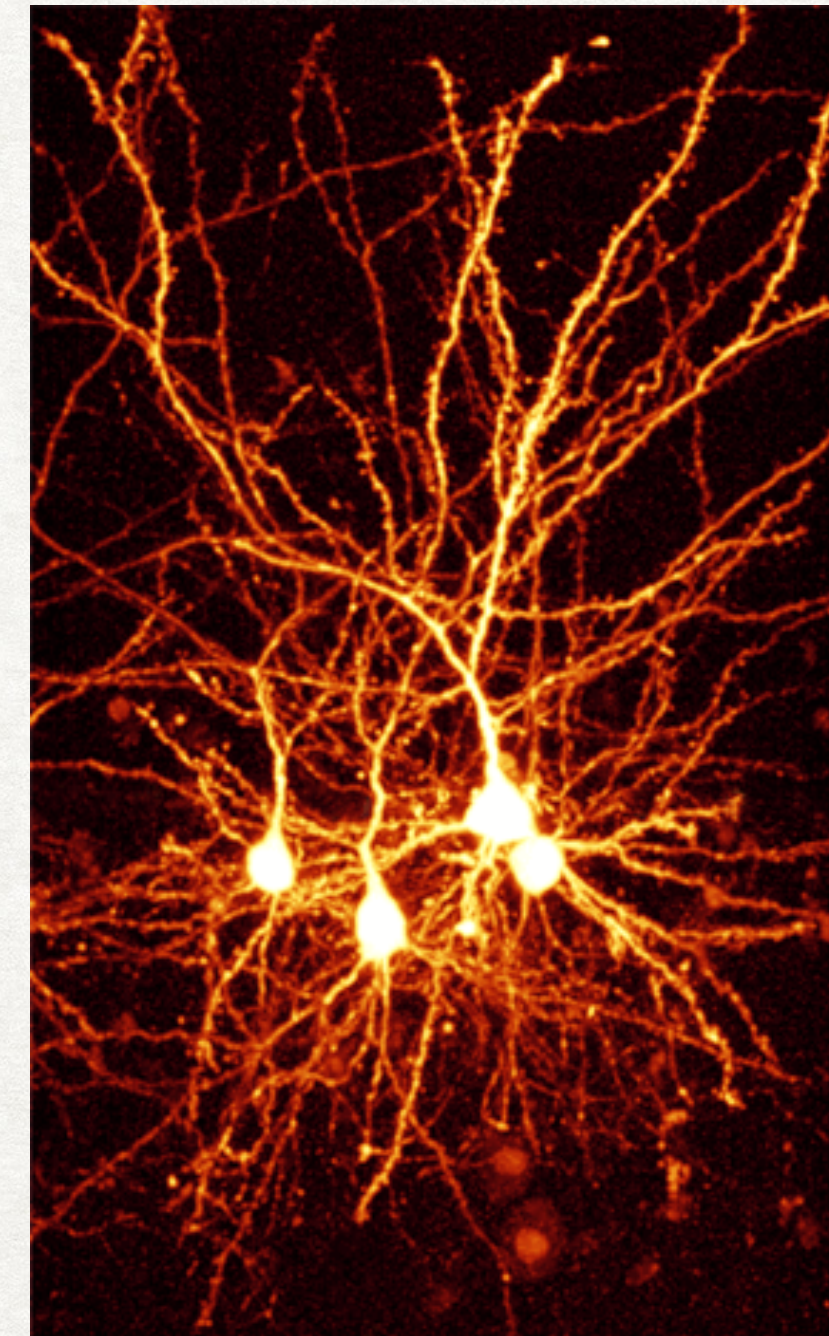
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MODELLING EXPERIMENTS

LEARNING FROM THE PERSPECTIVE OF NEUROSCIENCE

- Neurons
- Networks
- Modification of neural connection strength = Learning



Institute of Physiology
University of Berne

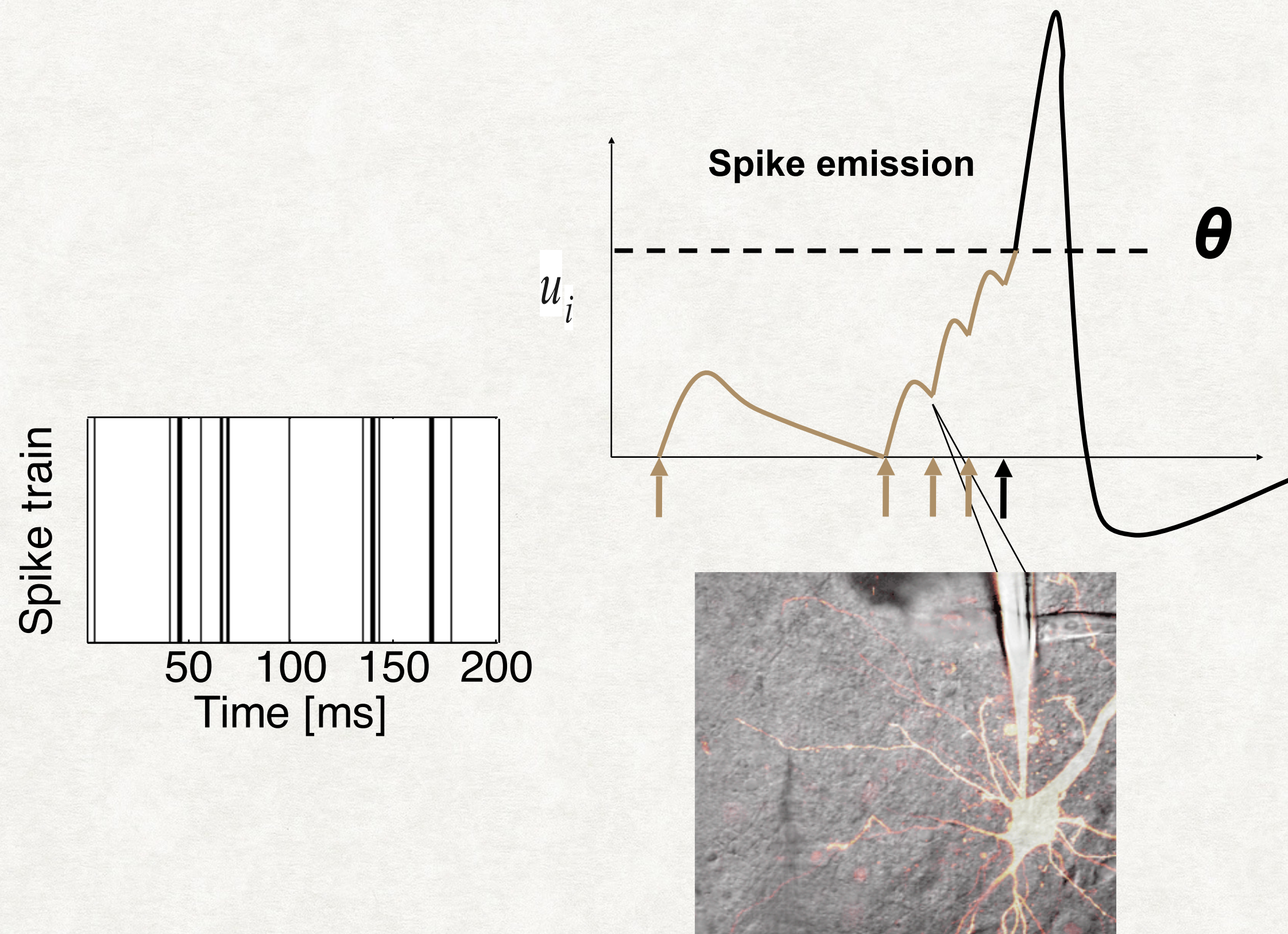


Ramon y Cahal

1mm³: 10,000 neurons 3km wires

MODELLING EXPERIMENTS

LEARNING FROM THE PERSPECTIVE OF NEUROSCIENCE



at time t_j^f : $\delta(t - t_j^f) \approx \frac{1}{\Delta t}$

$$\sum_{j,f} w_{ij}(t) \delta(t - t_j^f)$$

$$\tau \frac{du_i}{dt} = -u_i(t) + RI(t)$$

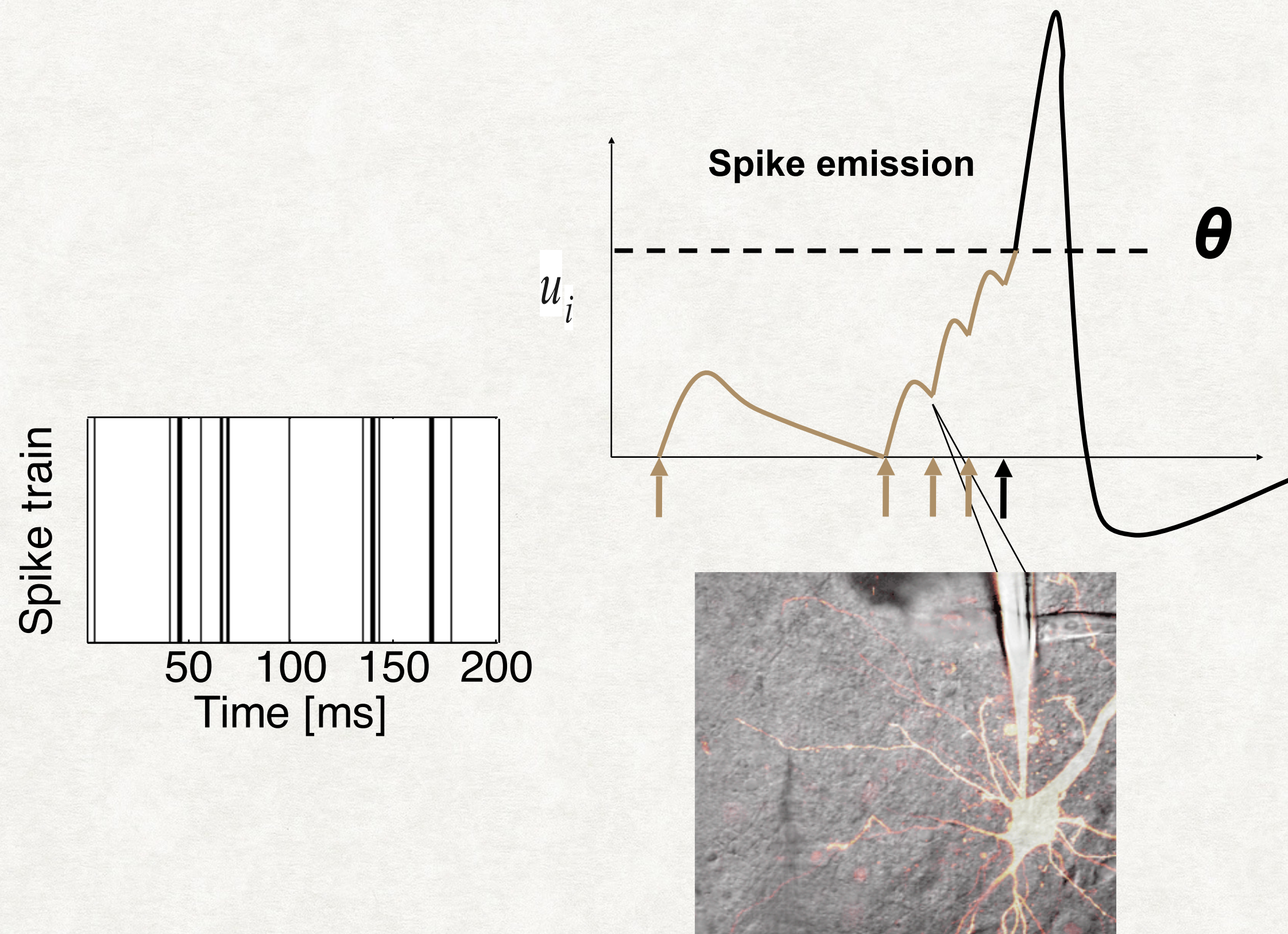
$u(t) > \theta$ emit spike

Stein, 1965
Gerstner et al, 2002

Can be formulated as probabilistic

MODELLING EXPERIMENTS

LEARNING FROM THE PERSPECTIVE OF NEUROSCIENCE



at time t_j^f : $\delta(t - t_j^f) \approx \frac{1}{\Delta t}$

$$\sum_{j,f} w_{ij}(t) \delta(t - t_j^f)$$

$$\tau \frac{du_i}{dt} = -u_i(t) + RI(t)$$

$$p_{\text{fire}} = \rho(t) \Delta t = \exp \left(\frac{u_t(t) - \theta}{\Delta u} \right) \Delta t$$

Stein, 1965
Gerstner et al, 2002

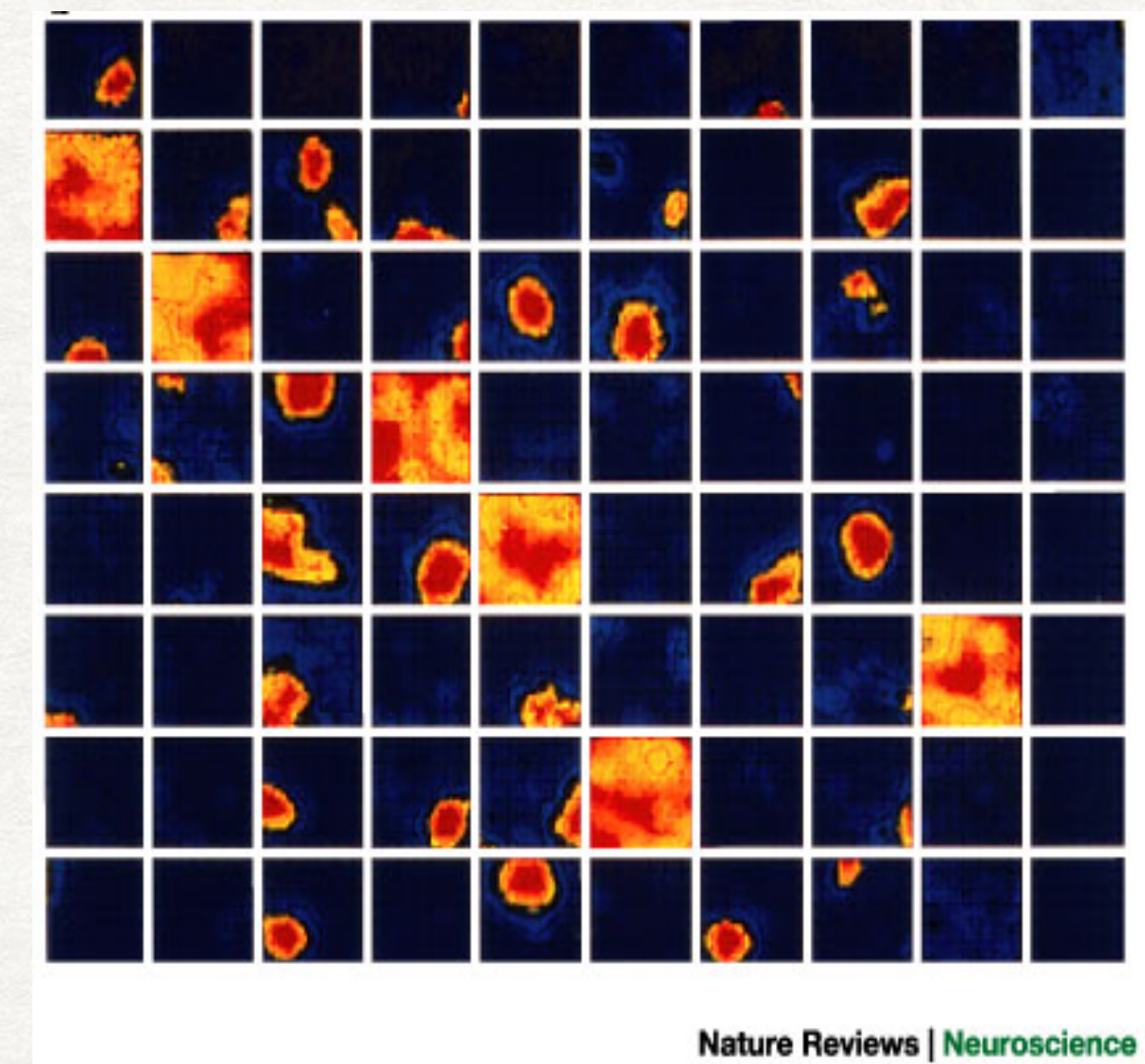
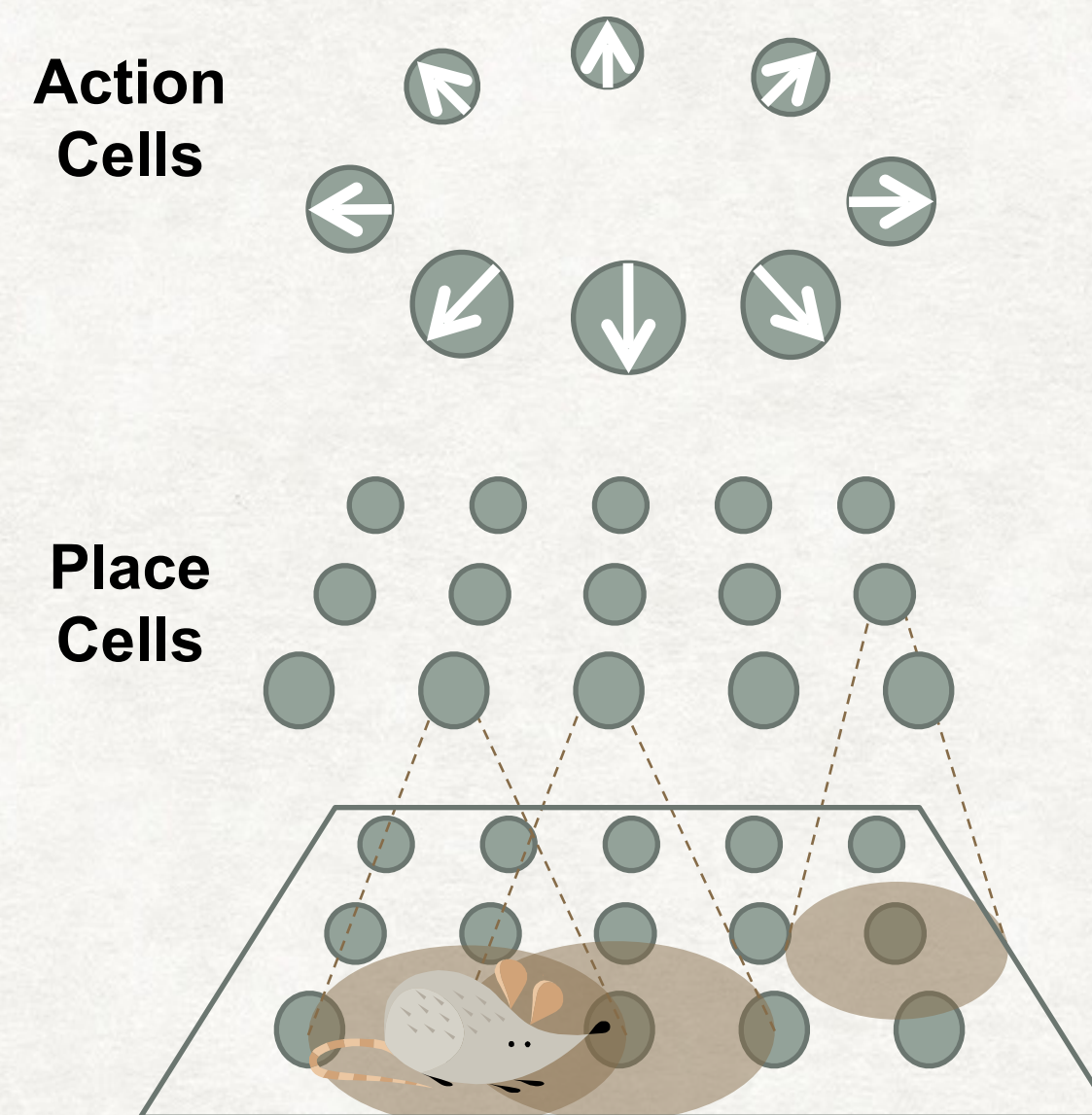
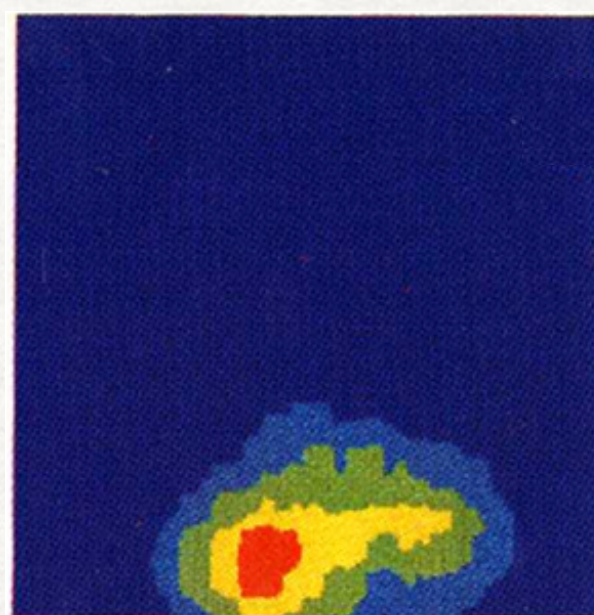
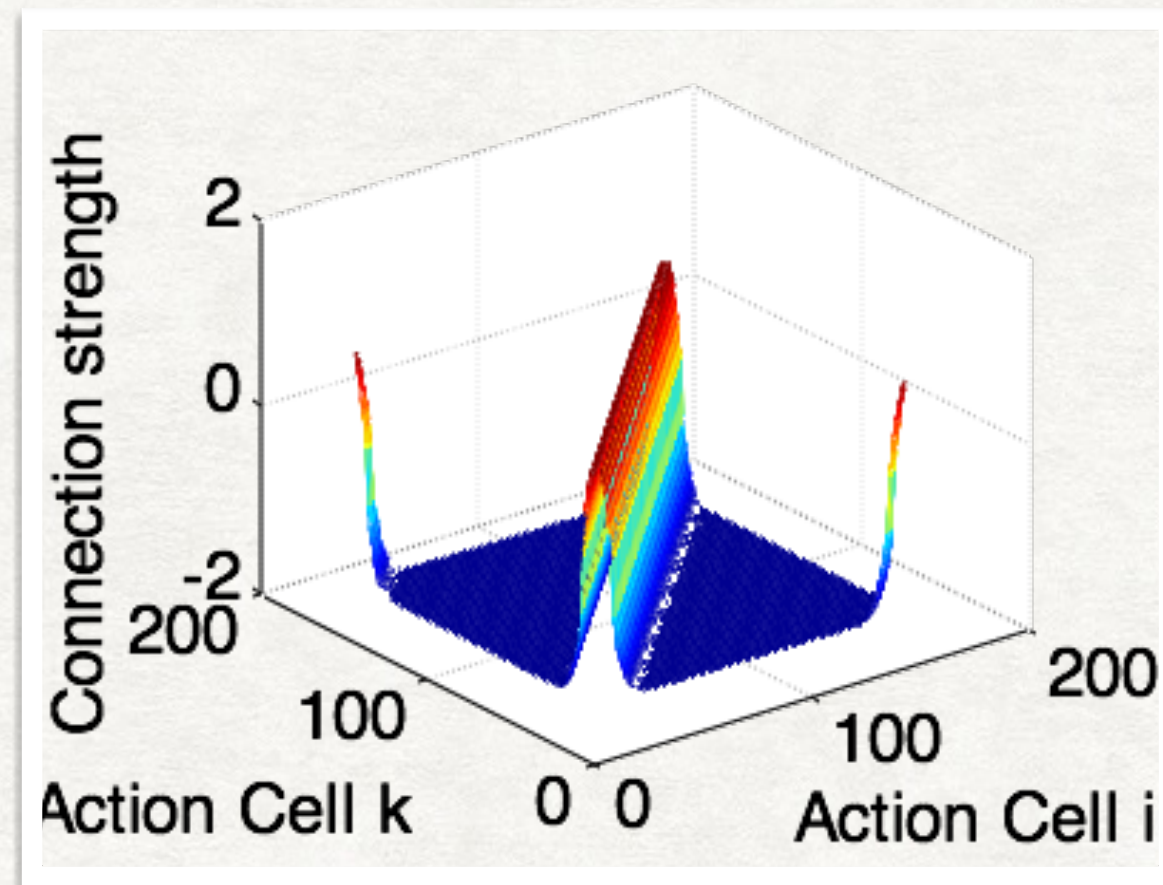
MODELLING EXPERIMENTS WITH REINFORCEMENT LEARNING

TALK OUTLINE

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MODELLING EXPERIMENTS

SPIKE-BASED REINFORCEMENT LEARNING OF RODENT NAVIGATION

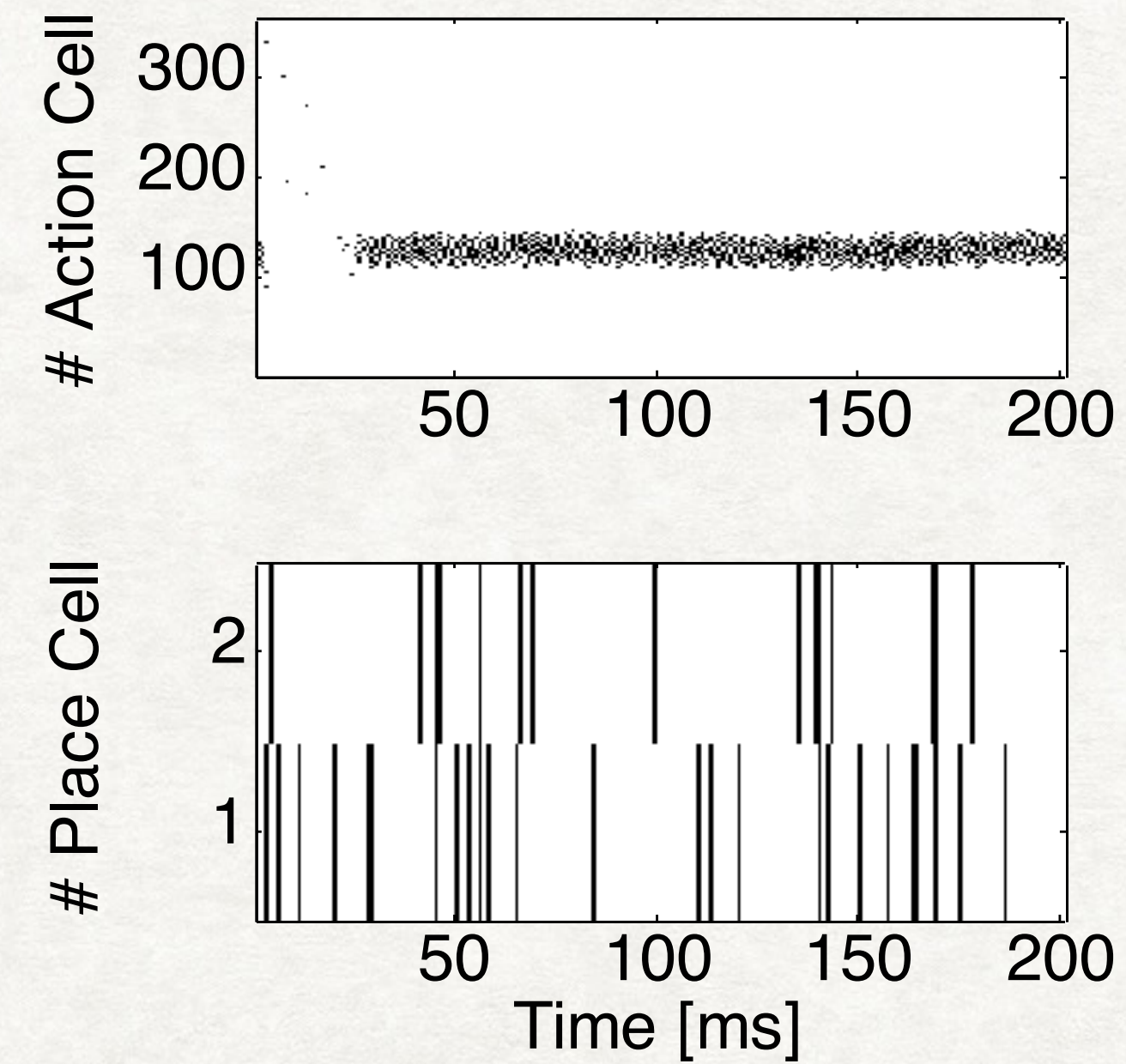
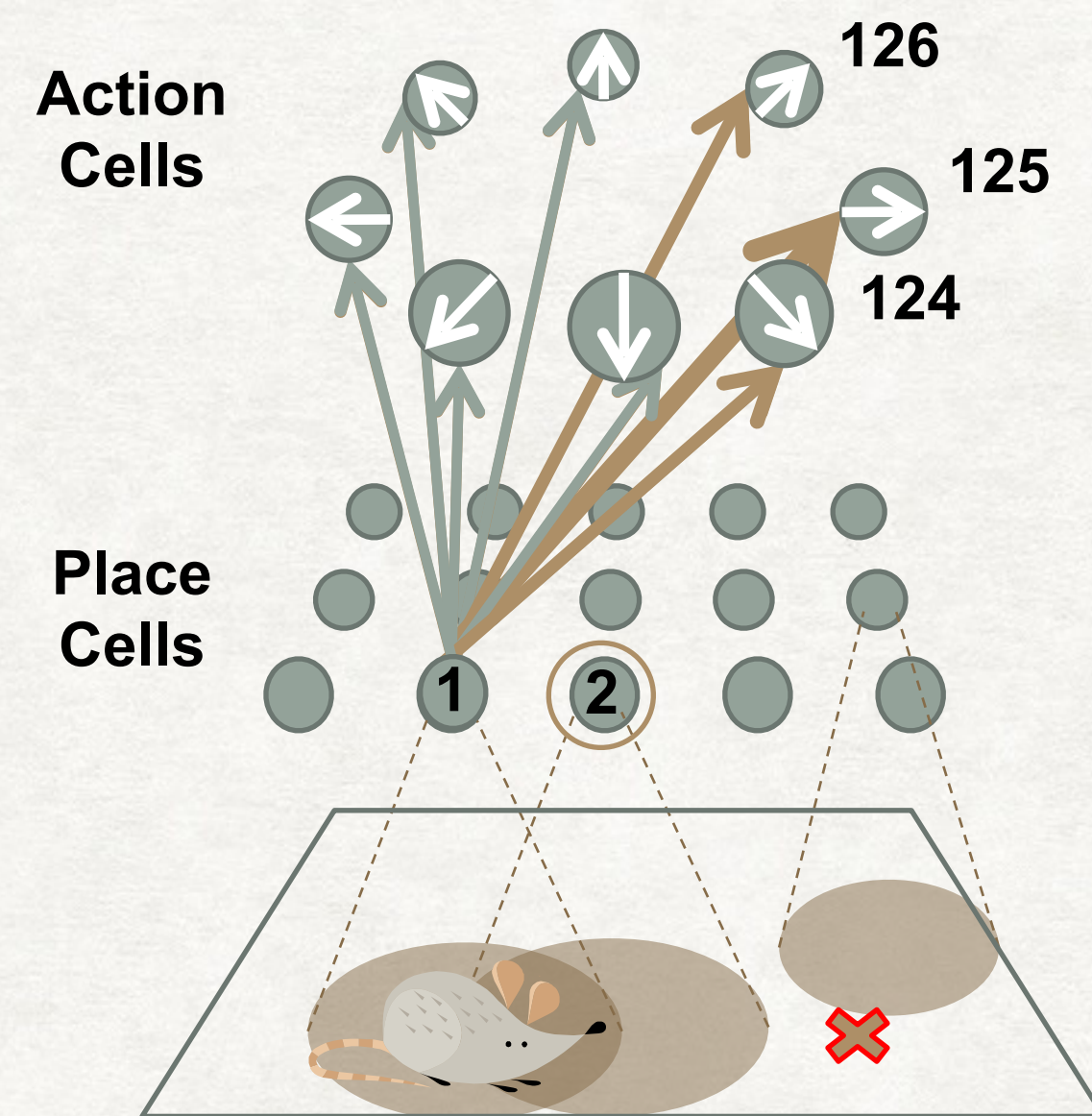


O'Keefe & Dostrovsky , 1971

Nakazawa et al 2004

MODELLING EXPERIMENTS

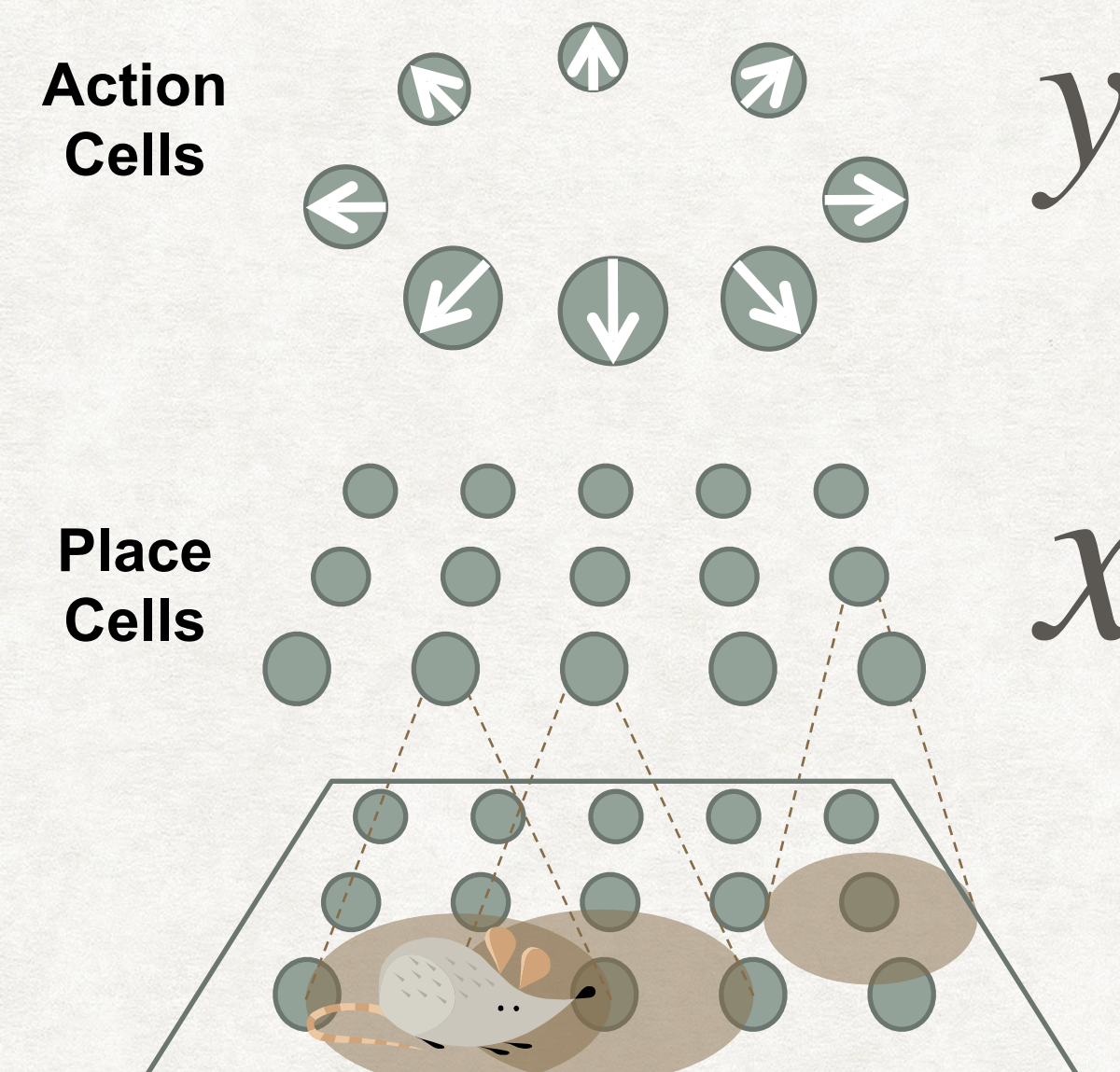
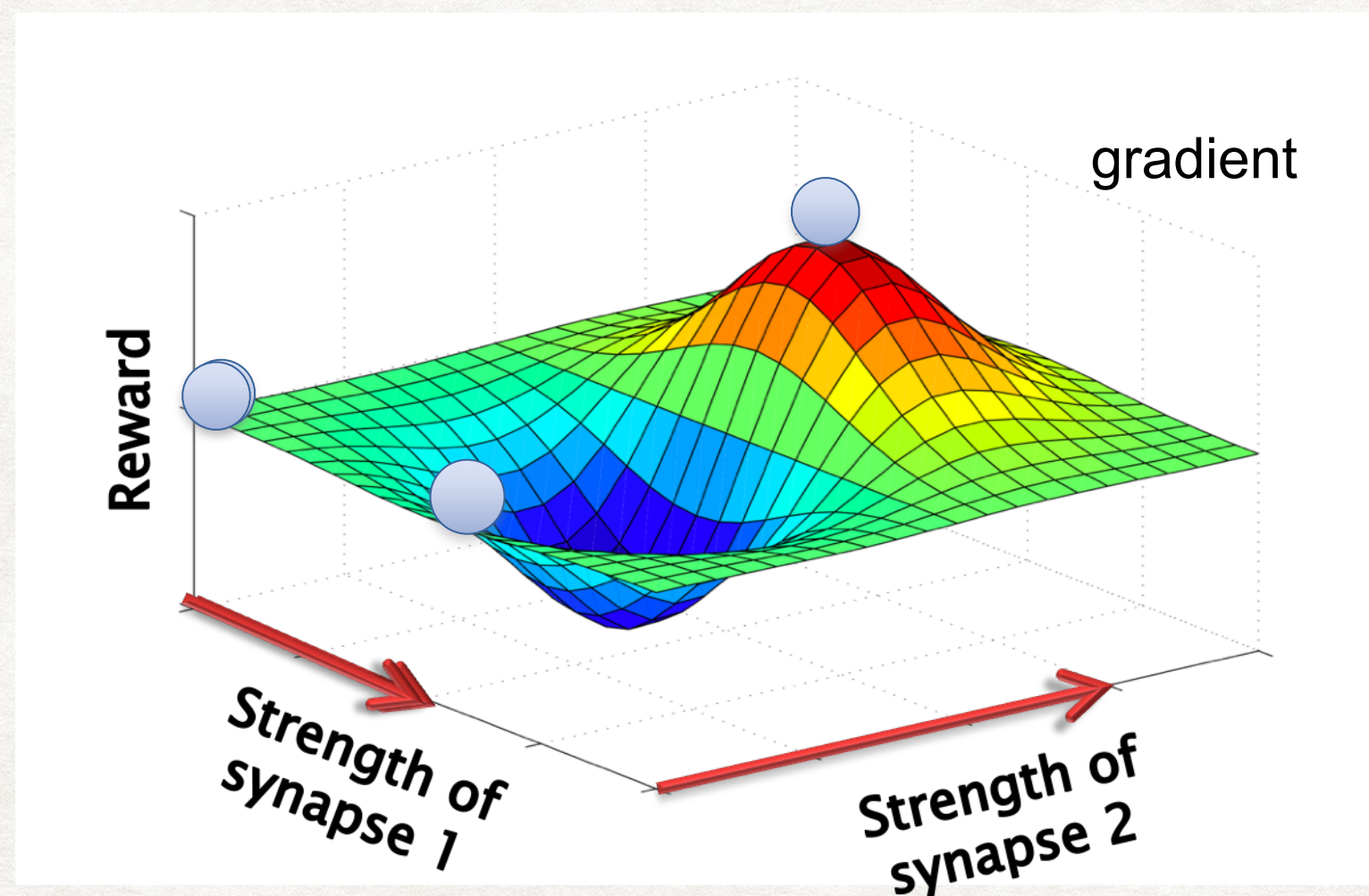
SPIKE-BASED REINFORCEMENT LEARNING OF RODENT NAVIGATION



MODELLING EXPERIMENTS

SPIKE-BASED REINFORCEMENT LEARNING OF RODENT NAVIGATION

$$\langle R \rangle_{x,y} = \sum_{x,y} R(x,y) P(y|x) P(x)$$



MODELLING EXPERIMENTS

SPIKE-BASED REINFORCEMENT LEARNING OF RODENT NAVIGATION

$$\langle R \rangle_{x,y} = \sum_{x,y} R(x,y) P(y|x) P(x)$$

Williams 1992
Xie and Seung, 2004
Pfister et al. 2006
Florian 2007
Vasilaki et al, 2009

$$\langle \Delta w \rangle_{x,y} = \alpha \frac{\partial \langle R \rangle_{x,y}}{\partial w}$$

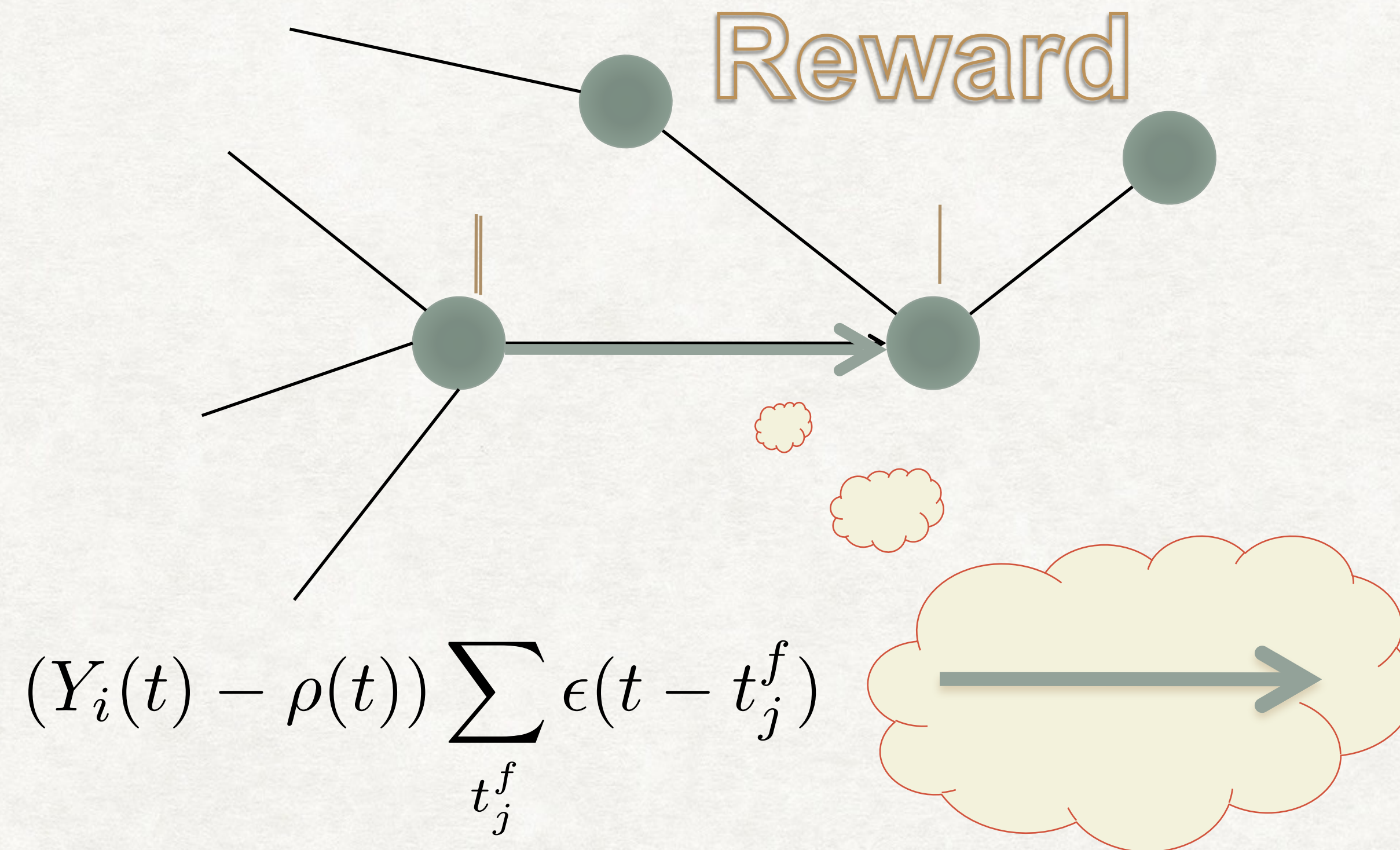
$$\frac{dw_{ij}}{dt} = \underbrace{\alpha}_{\text{Learning rate}} \underbrace{R(t)}_{\text{Reward}} \underbrace{e_{ij}(t)}_{\text{Eligibility trace}}$$

Instantaneous
probability of firing

$$\frac{de_{ij}}{dt} = -\underbrace{\frac{e_{ij}}{\tau_e}}_{\text{Time constant}} + \underbrace{(Y_i(t) - \rho(t))}_{\text{Spike}} \sum_{t_j^f} \underbrace{\epsilon(t - t_j^f)}_{\text{Input at synapse ij}}$$

MODELLING EXPERIMENTS

SPIKE-BASED REINFORCEMENT LEARNING OF RODENT NAVIGATION



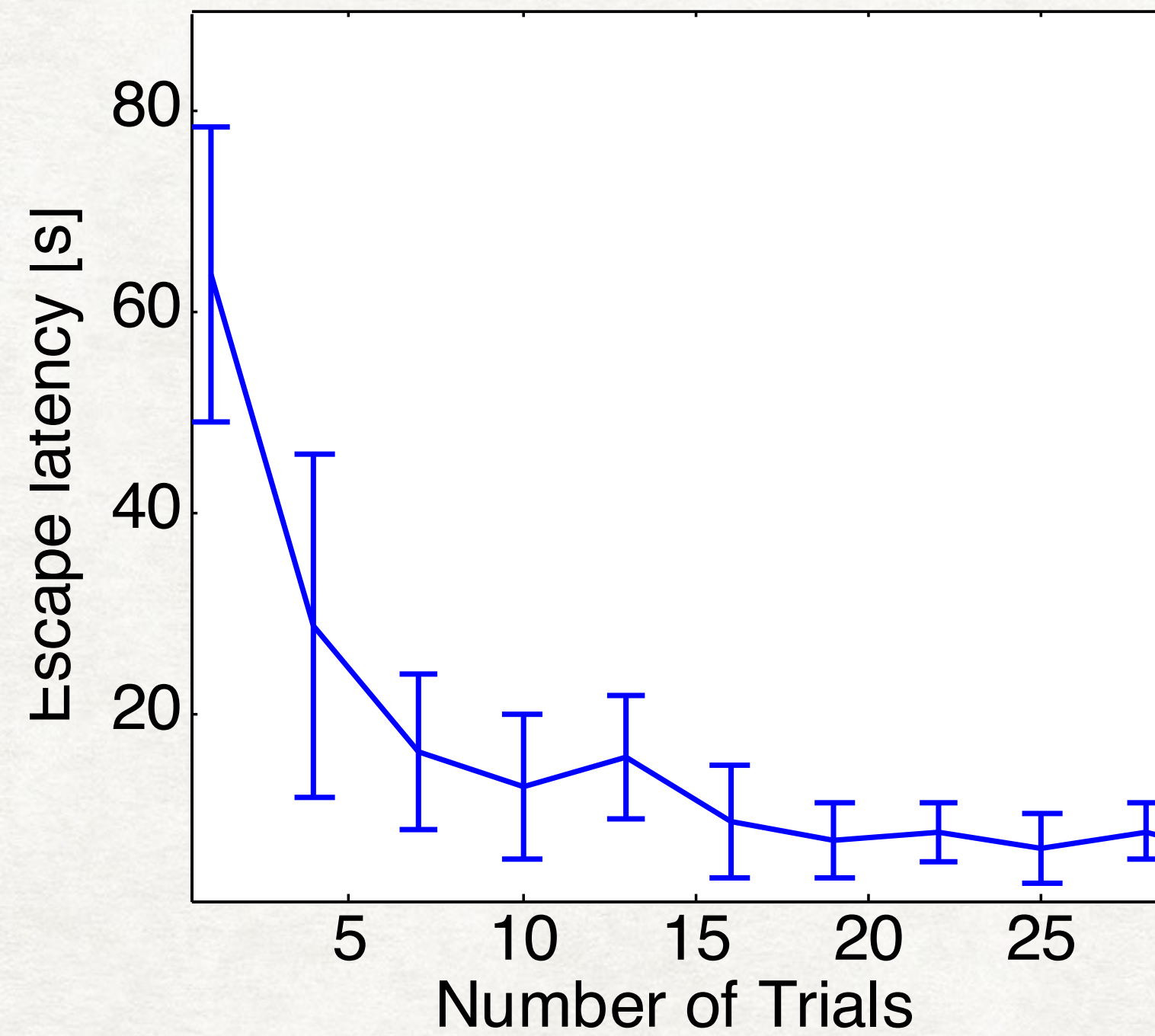
MODELLING EXPERIMENTS WITH REINFORCEMENT LEARNING

TALK OUTLINE

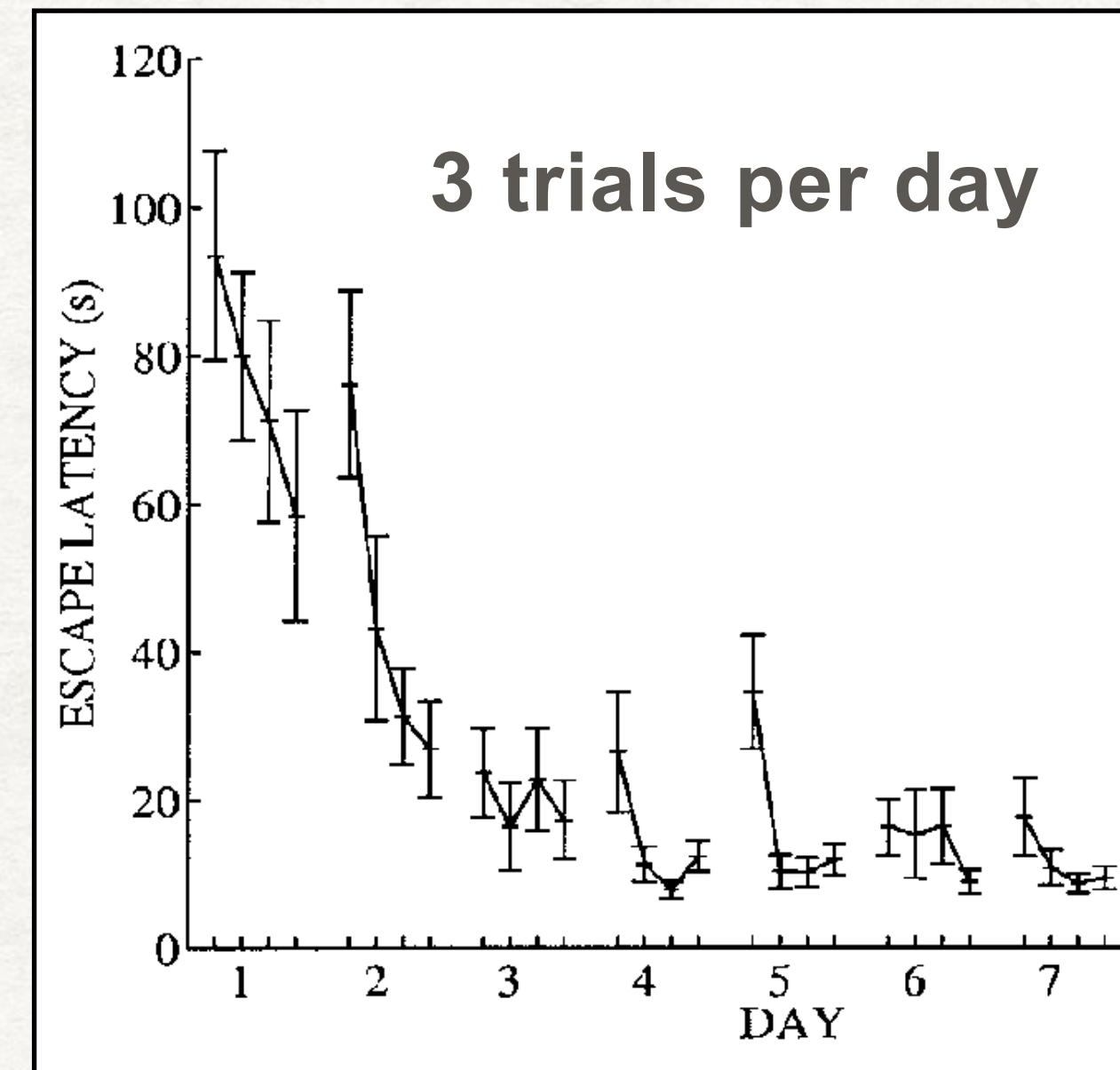
- “Vanilla” Temporal Difference Learning
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MODELLING EXPERIMENTS

RESULTS



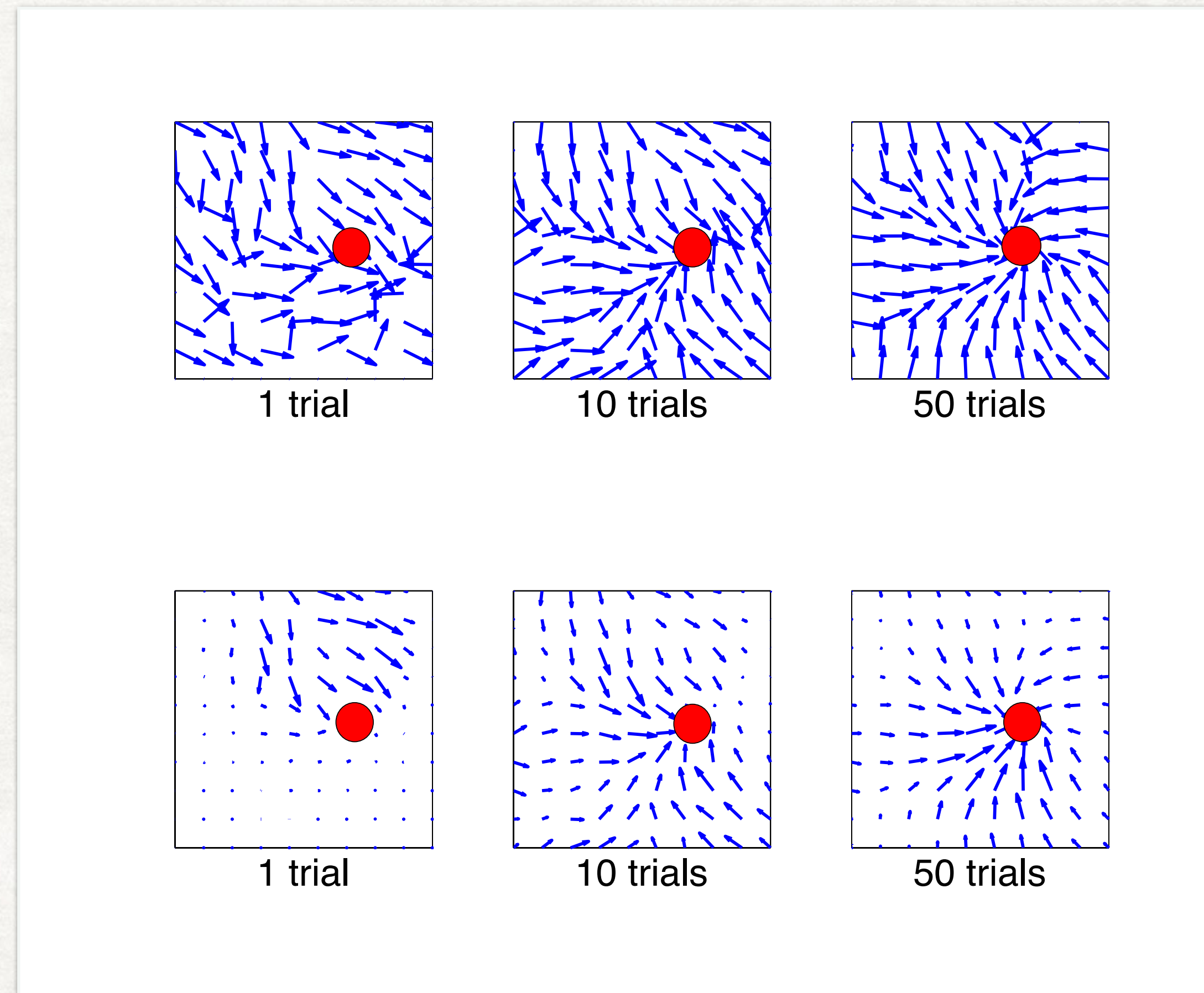
Simulations



Experiment

MODELLING EXPERIMENTS

RESULTS

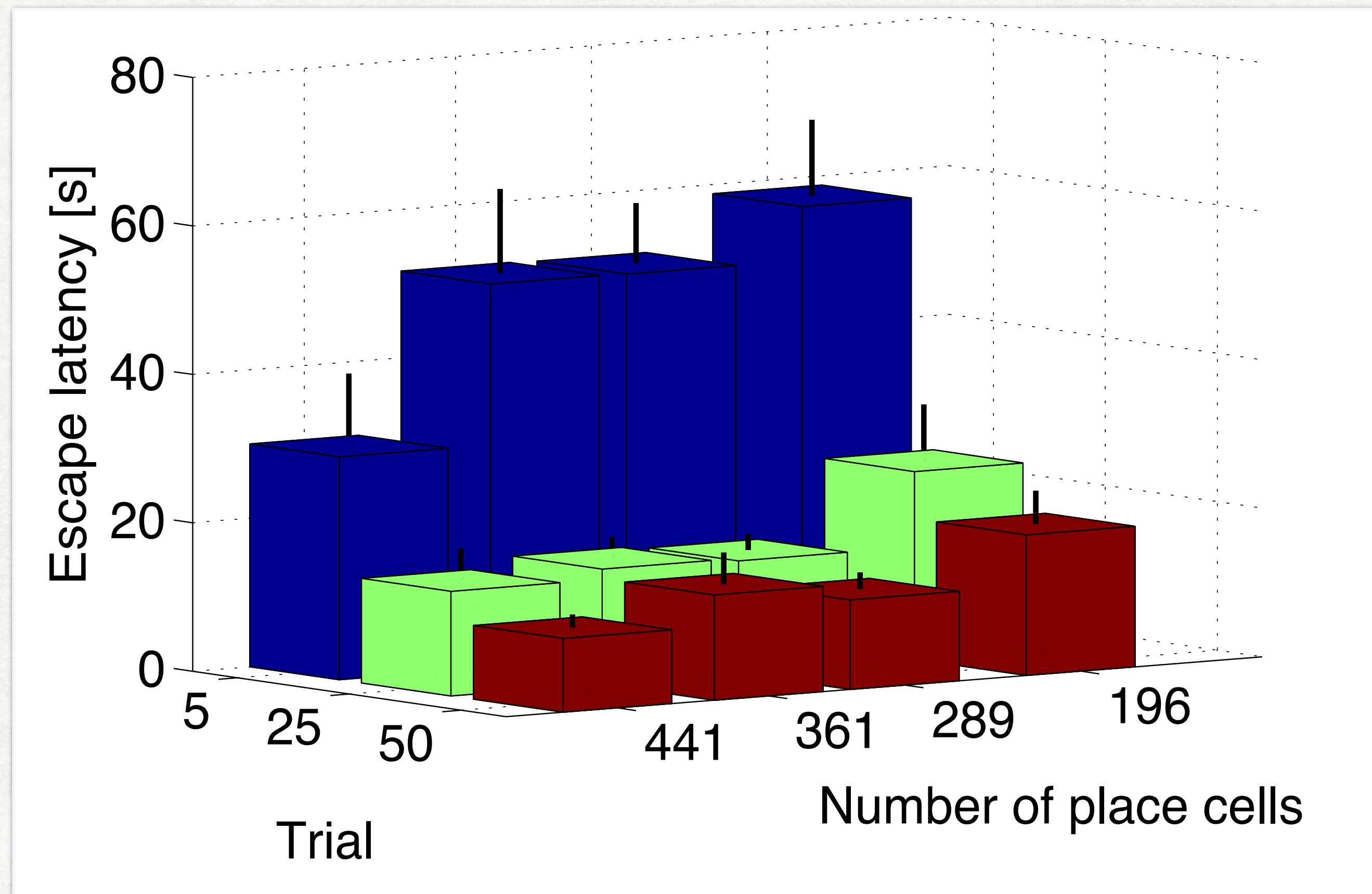


Normalised

Actual

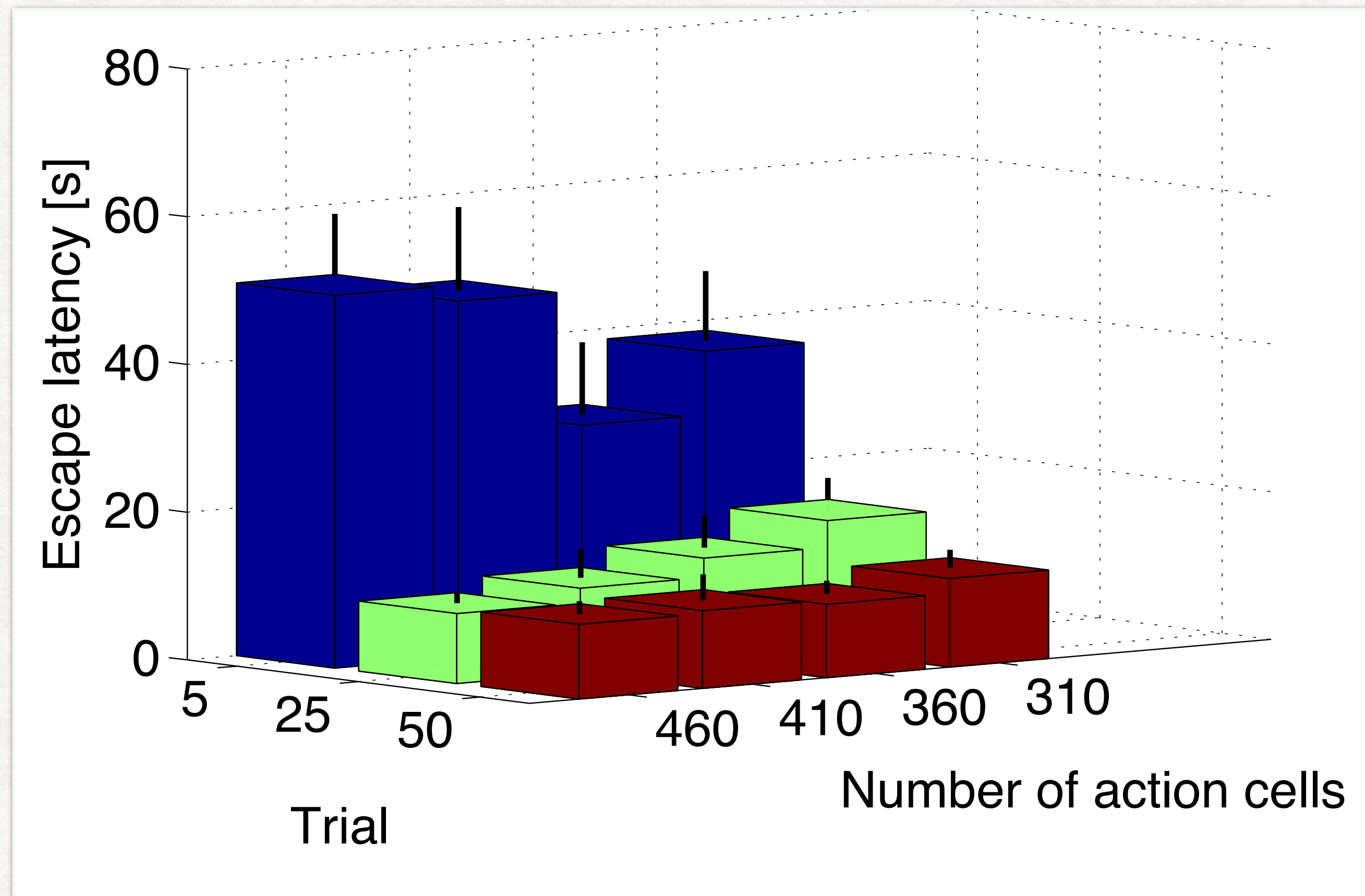
MODELLING EXPERIMENTS

RESULTS



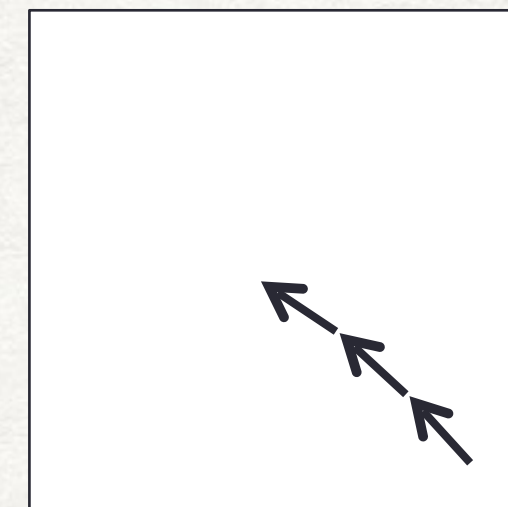
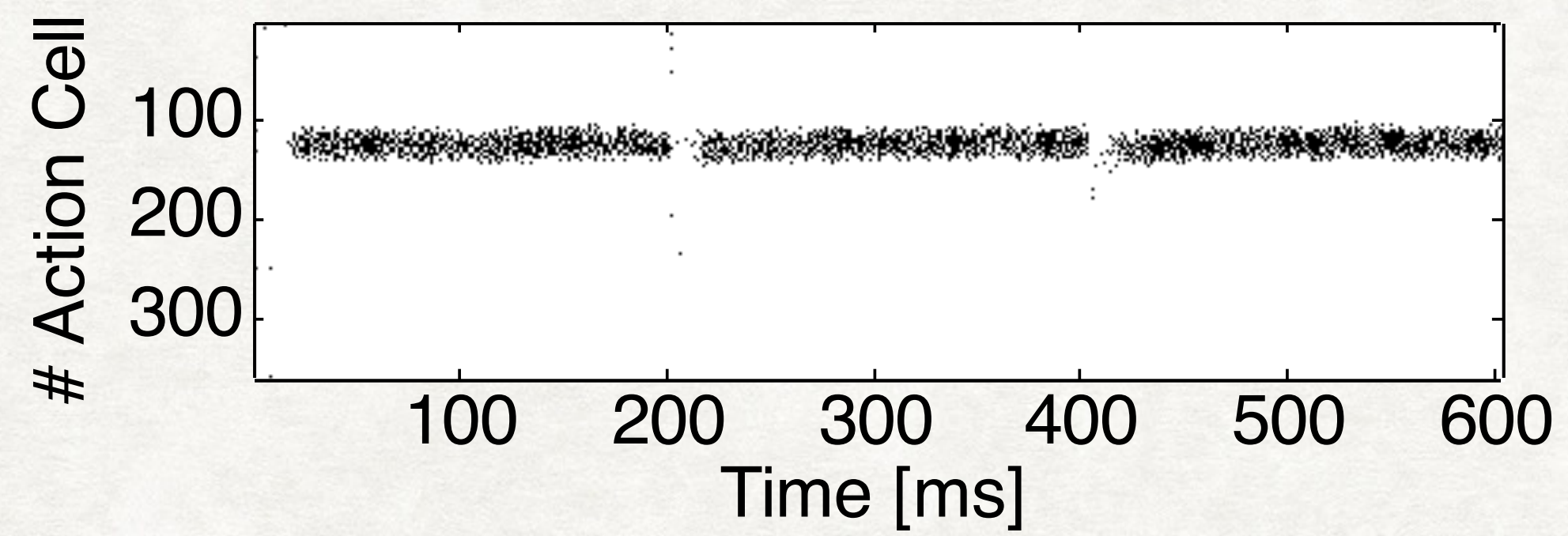
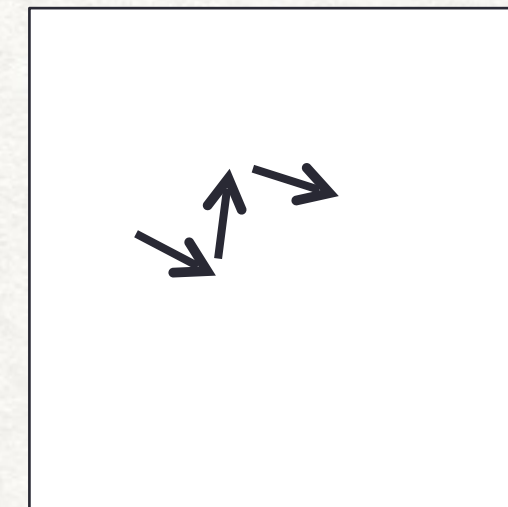
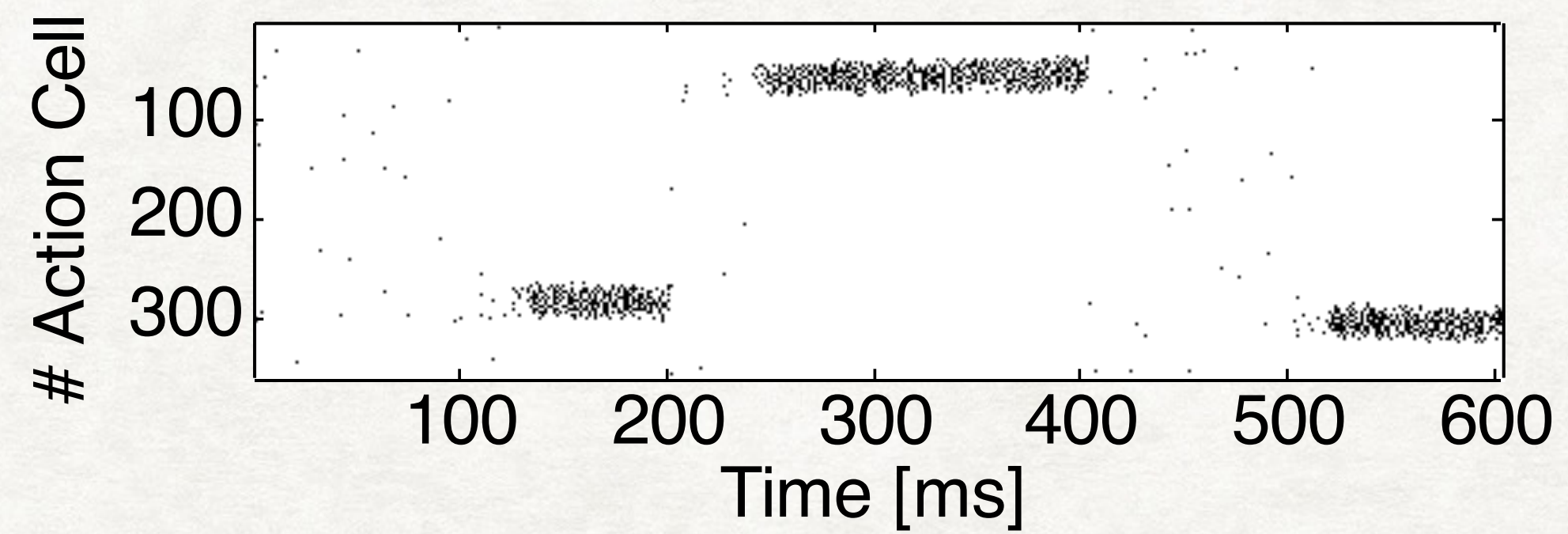
MODELLING EXPERIMENTS

RESULTS



MODELLING EXPERIMENTS

RESULTS



MODELLING EXPERIMENTS

RESULTS

OPEN  ACCESS Freely available online

PLOS COMPUTATIONAL BIOLOGY

Spike-Based Reinforcement Learning in Continuous State and Action Space: When Policy Gradient Methods Fail

Eleni Vasilaki^{1,2*}, Nicolas Frémaux¹, Robert Urbanczik³, Walter Senn³, Wulfram Gerstner¹

¹ Laboratory of Computational Neuroscience, EPFL, Lausanne, Switzerland, ² Department of Computer Science, University of Sheffield, Sheffield, United Kingdom,

³ Department of Physiology, University of Bern, Bern, Switzerland

MODELLING EXPERIMENTS

TAKE HOME MESSAGE

- Reinforcement Learning + Spiking Neurons.
- Similar performance to biological data.
- Good scaling properties due to continuous description of states and actions.
- Link to decision making.
- Can provide ideas for enriching the assignment.

THANK YOU!

